Strong and ϵ-Dependent Converses Based on Simple Changes of Measures Proofs

Mireille Sarkiss and Michèle Wigger

***Abstract*—The goal of this overview article is to present change of measure proof techniques that can establish various kinds of strong or probability-of-error-dependent converses (which we call ϵ-dependent converses). Compared to related previous articles, e.g., by Gu and Effros or by Tyagi and Watanabe, in this article we also show how change of measure proof techniques can establish converses under multiple constraints or expectation constraints. Specifically, we first prove the strong converse for the almost lossless source-coding problem with side-information, and show that the same change of measure can be applied to prove an ϵ-dependent converse under an expected-rate constraint instead of the standard maximum-rate constraint. We further present a change of measure proof for the lossy source-coding problem with side-information under multiple distortion constraints and expected-rate or maximum-rate constraints. Under multiple distortion constraints and maximum rate-constraints the minimum compression rate coincides with the minimum rate under vanishing error-probability only when the sum of error probabilities does not exceed 1, after which a phase transition happens. Under expected-rate constraints the minimum compression rate depends on all error-probabilities and we provide an exact converse result. The paper also proves a strong converse for distributed hypothesis testing, for channel coding, and for an integrated sensing and communication system which combines channel coding and hypothesis testing.**

***Index Terms*—Converse proof, change of measure, source coding, distributed hypothesis testing, channel coding, integrated sensing and communication.**

I. Introduction

Change of measure proof techniques have a rich history in information theory. The idea is to change the distribution of underlying sources or channels to obtain a simpler problem, for example without errors, solve the simpler problem, and prove relations between the original and the new problem.

This overview article builds on the converse proofs and results in [1]–[6]. These proofs build on the change of measure converse proofs by Gu and Effros [7], [8] (which treat certain classes of source coding networks) and by Tyagi and Watanabe [9] (which treat source coding, channel coding, and function computation problems).

Specifically, our proof method has three main components:

1) We define the set $\mathcal{D}_n$ of jointly- or conditionally-typical source or channel realizations for which there are no errors, and we obtain the new measure by restricting the original measure to this new set.
2) We bound the rates, distortions, or detection error exponents for the new problem, also introducing auxiliary random variables, and (if necessary) show that the solution of the new problem is not far from the solution to the original one.
3) In our final steps we prove that the newly introduced auxiliary random variables satisfy certain asymptotic Markov chains, and normalized multi-letter entropy quantities on the new measure converge to their counterparts under the original distribution.

The proof steps performed under item 3 are generally trivial under the original measure. In this sense, the transformation of the problem into a zero-error problem is the price to pay for the applied change of measure. On the other hand, the change of measure makes item 2 amenable for proof.

As already mentioned, the overall approach of our proof technique follows the techniques by Gu and Effros in [7], [8] and by Tyagi and Watanabe in [9]. In particular, items 1 and 2 are mostly taken from the Tyagi-Watanabe proof in [9], except that in their proof, the set $\mathcal{D}_n$ is not a subset of the jointly- or conditionally-typical set. As a consequence, they apply a different item 3 in their proof, proving a variational characterizations of the multi-letter problem, single-letterizing this variational characterization, and then showing that the result corresponds to the variational characterization of the desired single-letter expression. Notice that for most examined examples our proofs of the asymptotic Markov chains coincide with some of the steps in the single-letterization of the Tyagi-Watanabe proof, and thus do not add additional complexity to our proof.

Gu and Effros [8] in contrast identify the same sets $\mathcal{D}_n$ as ours, but their new measure concentrates on $\mathcal{D}_n$ only in the asymptotic limit as the blocklength $n \to \infty$. As a consequence, the new measure in [8] does not have zero probability of error, and one has to show that the error probability vanishes for large blocklengths.

Besides these rather minor technical differences, our article also deviates from [7], [8] and [9] in the choice of the considered scenarios. Besides illustrating change-of-measure converse proofs for different applications (source coding, channel coding, distributed hypothesis testing), in this overview article we particularly focus on setups with multiple constraints (multiple distortion constraints or simultaneous decoding and detection constraints) or with expectation constraints. The goal is to illustrate that the change-of-measure converse proof technique initiated in [7], [8] and [9] is a great tool also to address these setups.

We begin this article (Section III) by presenting our proof technique for the lossless source coding problem with side-information under a maximum-rate constraint or an expected-rate constraint. Under a maximum-rate constraint the strong converse holds and the minimum compression rate coincides with the minimum compression rate under a vanishing error probability. Under an expected-rate constraint the minimum compression rate scales with $(1-\epsilon)$ for ϵ denoting the allowed

M. Sarkiss is with SAMOVAR, Télécom SudParis, Institut Polytechnique de Paris, 91120 Palaiseau, France, mireille.sarkiss@telecom-sudparis.eu.

M. Wigger is with LTCI, Télécom Paris, Institut Polytechnique de Paris, 91120 Palaiseau, France, michele.wigger@telecom-paris.fr.

error probability. For the lossless source coding setup our proof is particularly simple, even under an expected-rate constraint. In fact, the same change of measure (the same set $\mathcal{D}_n$) can be used for both proofs.

We next (Section IV) consider the *lossy* source coding problem with side-information under one or two distortion constraints and under maximum- or expected-rate constraints. Again the same change of measure can be applied irrespective of whether the maximum-rate or the expected-rate are constrained. Under two distortion constraints and a maximum-rate constraint (Subsection IV-C) the strong converse (i.e., that the minimum compression rate coincides with the minimum rate under vanishing error probabilities) holds only when the sum of the two error probabilities does not exceed 1. Otherwise a simple randomized strategy can achieve a smaller compression rate.

Under two distortion constraints and an expected-rate constraint (see Subsection IV-D) the desired converse result and proof are slightly more involved. In particular, the proof requires three changes of measure applied in parallel on disjoint subsets $\mathcal{D}_{1,n}, \mathcal{D}_{2,n}, \mathcal{D}_{12,n}$, and the final result combines two of the expressions encountered under the maximum error probability.

The latter Sections V–VII of this article focus on binary hypothesis testing and channel coding. We again start with a simple warm-up example: Stein's asymmetric binary hypothesis testing setup with a single sensor. Then we present our change of measure proof technique for the distributed hypothesis testing scenario when testing is against independence. A change of measure proof technique is also proposed for channel coding and for an integrated sensing and communication (ISAC) setup which combines channel coding and hypothesis testing. For channel coding the proof presented in this paper is probably more complicated than other related proofs in the literature. Its interest however comes from its suitability to treat multi-objective scenarios where data communication is combined with other purposes. ISAC is a timely example of a such multi-objective scenario, in which an additional radar receiver (or the transmitter) wishes to detect the distribution underlying the channel.

Our theorems in this article only state converse results. They are however all exact in the sense that matching achievability proofs are either well known or can be obtained using simple and standard techniques. Moreover, we only present converse results for fixed error probabilities $\epsilon \in [0,1)$ (or $\delta \in [0,1)$). A quick inspection of the proofs shows that all our results hold also for error probabilities ϵ_n that converge to 1 sub-exponentially fast in the blocklength, i.e., so that the following holds:

$$\underline{\lim}_{n\to\infty} \frac{1}{n} \log(1-\epsilon_n) = 0. \tag{1}$$

The change of measure converse proof techniques discussed in this article thus can be used to prove so called exponentially-strong converses. They however seem insufficient to achieve the optimal error exponents. We however believe that the strength of our proof technique lies in its simplicity and similarity to known weak converse proofs (see also the discussion in [9] regarding their change of measure proof technique), and its wide applicability especially also to scenarios with multiple constraints and expectation constraints.

A. Notation

We mostly follow standard notation. In particular, we abbreviate $(x_1, \dots, x_n)$ by x^n and $(x_{t+1}, \dots, x_n)$ by x_{t+1}^n. We further abbreviate *independent and identically distributed* as *i.i.d.* and *probability mass function* as *pmf*. Also, we denote by $\pi_{x^n y^n}$ the joint type of the sequences (x^n, y^n):

$$\pi_{x^n y^n}(a,b) \triangleq \frac{n_{x^n,y^n}(a,b)}{n}, \tag{2}$$

and we use $\mathcal{T}_\mu^{(n)}(P_{XY})$ to denote the jointly strongly-typical set as in [10, Definition 2.9]. Finally, we use Landau notation $o(1)$ to indicate any function that tends to 0 for blocklengths $n \to \infty$.

Throughout this manuscript, $\{\mu_n\}_{n=1}^\infty$ is a sequence of small positive numbers satisfying[1],

$$\lim_{n\to\infty} \mu_n = 0 \tag{3a}$$

$$\lim_{n\to\infty} n \cdot \mu_n^2 = \infty. \tag{3b}$$

II. Auxiliary Lemmas

Let $\{(X_i, Y_i)\}_{i=1}^\infty$ be a sequence of pairs of i.i.d. random variables according to the pmf P_{XY}. For each positive integer n let $\mathcal{D}_n$ be a subset of the strongly-typical set $\mathcal{T}_{\mu_n}^{(n)}(P_{XY})$ so that its probability

$$\Delta_n := \Pr[(X^n, Y^n) \in \mathcal{D}_n] \tag{4}$$

satisfies

$$\lim_{n\to\infty} \frac{1}{n} \log \Delta_n = 0. \tag{5}$$

Let further $(\tilde{X}^n, \tilde{Y}^n)$ be random variables of joint pmf

$$P_{\tilde{X}^n\tilde{Y}^n}(x^n, y^n) = \frac{P_{XY}^{\otimes n}(x^n, y^n)}{\Delta_n} \cdot \mathbb{1}\{(x^n, y^n) \in \mathcal{D}_n\} \tag{6}$$

and T be uniform over $\{1, \dots, n\}$ independent of all other random variables.

Lemma 1: For the distribution in (6), the following limits hold as $n \to \infty$. For each pair $(x,y) \in \mathcal{X} \times \mathcal{Y}$.

$$P_{\tilde{X}_T\tilde{Y}_T}(x,y) \quad \to \quad P_{XY}(x,y), \tag{7}$$

and

$$\lim_{n\to\infty} \frac{1}{n} H(\tilde{X}^n, \tilde{Y}^n) = \lim_{n\to\infty} H(\tilde{X}_T, \tilde{Y}_T) \quad = \quad H(X,Y) \tag{8}$$

$$\lim_{n\to\infty} \frac{1}{n} H(\tilde{Y}^n) = \lim_{n\to\infty} H(\tilde{Y}_T) \quad = \quad H(Y) \tag{9}$$

$$\lim_{n\to\infty} \frac{1}{n} H(\tilde{X}^n | \tilde{Y}^n) = \lim_{n\to\infty} H(\tilde{X}_T | \tilde{Y}_T) \quad = \quad H(X|Y). \tag{10}$$

Moreover, over an arbitrary pmf Q_{XY} over $\mathcal{X} \times \mathcal{Y}$:

$$\lim_{n\to\infty} \frac{1}{n} D(P_{\tilde{X}^n\tilde{Y}^n} \| Q_{XY}^{\otimes n}) \quad = \quad D(P_{XY} \| Q_{XY}). \tag{11}$$

Proof: See Appendix A. ■

Above lemma suffices to establish strong and ϵ-dependent converses for the Slepian-Wolf lossless source coding problem with side-information [11], [12], see Section III.

[1] Condition (3b) ensures that the probability of the strongly typical set $\mathcal{T}_{\mu_n}^{(n)}(P_{XY})$ under $P_{XY}^{\otimes n}$ tends to 1 as $n \to \infty$ [10, Remark to Lemma 2.12].

To establish the other results, we will need an additional lemma involving an auxiliary random variable.

Lemma 2: Let $(\tilde{X}^n, \tilde{Y}^n)$ and T be defined as above and define $\tilde{\mathsf{M}}$ as a function of $\tilde{X}^n$ and

$$U \triangleq (\tilde{X}^{T-1}, \tilde{Y}_{T+1}^n, \tilde{\mathsf{M}}, T). \tag{12}$$

Then:

$$\lim_{n\to\infty} \left[\frac{1}{n} H(\tilde{X}^n|\tilde{Y}^n, \tilde{\mathsf{M}}) - H(\tilde{X}_T|\tilde{Y}_T, U)\right] = 0 \tag{13}$$

and

$$\frac{1}{n} H(\tilde{Y}^n|\tilde{X}^n, \tilde{\mathsf{M}}) \leq H(\tilde{Y}_T|\tilde{X}_T, U). \tag{14}$$

Proof: See Appendix B. ■

III. Lossless Compression with Side-Information

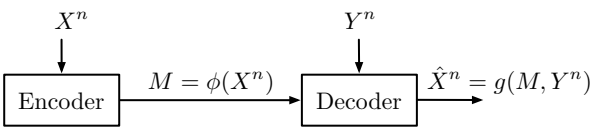

Fig. 1: Almost lossless source coding of a DMS.

Consider the two-terminal lossless source coding setup [11] with correlated side-information in Figure 1 where

$$(X^n, Y^n) \text{ i.i.d. } \sim P_{XY}, \tag{15}$$

for a given pmf P_{XY} on a finite product alphabet $\mathcal{X} \times \mathcal{Y}$. The encoder produces the message

$$\mathsf{M} = \phi^{(n)}(X^n) \tag{16}$$

for some encoding function $\phi^{(n)}(\cdot)$ and the decoder produces the reconstruction sequence

$$\hat{X}^n = g^{(n)}(\mathsf{M}, Y^n) \tag{17}$$

over some finite reconstruction alphabet $\mathcal{X}^n$ and a given reconstruction function $g^{(n)}(\cdot)$.

The message is typically constrained by a *maximum-rate constraint*:

$$\mathsf{M} \in \{1, \ldots, 2^{nR}\} \tag{18}$$

or equivalently, we assume that $\mathsf{M} \in \{0,1\}^\star$ is a bitstring and we constrain its length to nR bits:

$$\text{len}(\mathsf{M}) \leq nR. \tag{19}$$

Alternatively, we will also impose an *expected-rate constraint*, in which case $\mathsf{M} \in \{0,1\}^\star$ is necessarily a bit-string and has to satisfy

$$\mathbb{E}[\text{len}(\mathsf{M})] \leq nR_{\text{avg}}. \tag{20}$$

Notice that under the expected rate-constraint (20) variable-length coding can be advantageous whereas under the maximum-rate constraint (19) variable-length coding gives no benefits in achievable rates as we define them in Definition 1.

In the following subsections we present and prove converse results on the minimum compression rate under maximum- or expected-rate constraints. Under a maximum-rate constraint, the minimum compression rate was first established by Slepian and Wolf [11] for a vanishing error probabilities and by Oohama and Han for positive error probabilities. (In fact, [12] proves an exponentially-strong converse.) Without side-information, the strong converse under the maximum-rate constraint follows immediately from the results by [13], [14] and the exponentially-strong converse was proved in [15].

A. Strong Converse under a Maximum-Rate Constraint

Consider the maximum-rate constraint (19).

Definition 1: Given a permissible error probability $\epsilon \in [0,1)$. A *maximum rate* $R > 0$ is ϵ-achievable if there exists a sequence of encoding and decoding functions $\{(\phi^{(n)}, g^{(n)})\}_{n=1}^\infty$ satisfying (19) and

$$\overline{\lim_{n\to\infty}} \Pr\left[X^n \neq g^{(n)}\left(\phi^{(n)}(X^n)\right)\right] \leq \epsilon. \tag{21}$$

Theorem 1 ([12]): For any $\epsilon \in [0,1)$, any maximum rate $R < H(X|Y)$ is not ϵ-achievable.

In other words, if one only allows for a maximum rate $R < H(X|Y)$, then for any sequence of encoding and reconstruction functions $\phi^{(n)}$ and $g^{(n)}$, the probability of *correct reconstruction* vanishes for increasing blocklengths.

Proof: Fix a sequence $\{(\phi^{(n)}, g^{(n)})\}_{n=1}^\infty$ satisfying (19) and (21).

<u>*Change of Measure:*</u> Define for each n the subset

$$\mathcal{D}_n := \left\{(x^n, y^n) \in \mathcal{T}_{\mu_n}^{(n)}(P_{XY}) \colon g^{(n)}\left(\phi^{(n)}(x^n), y^n\right) = x^n\right\}, \tag{22}$$

i.e., the set of all typical (x^n, y^n)-sequences with no reconstruction error $\hat{X}^n$. Denote the probability of this set by

$$\Delta_n \triangleq \Pr[(X^n, Y^n) \in \mathcal{D}_n], \tag{23}$$

and notice that by the union bound

$$\begin{aligned}\Delta_n &= 1 - \Pr\Big[(X^n, Y^n) \notin \mathcal{T}_{\mu_n}^{(n)}(P_{XY}) \text{ or } \\ &\qquad g^{(n)}\left(\phi^{(n)}(X^n), Y^n\right) \neq X^n\Big] \qquad (24)\\ &\geq 1 - \Pr\left[(X^n, Y^n) \notin \mathcal{T}_{\mu_n}^{(n)}(P_{XY})\right] \\ &\quad - \Pr\left[g^{(n)}\left(\phi^{(n)}(X^n), Y^n\right) \neq X^n\right] \qquad (25)\\ &\geq 1 - \frac{|\mathcal{X}||\mathcal{Y}|}{4\mu_n^2 n} - \epsilon_n, \qquad (26)\end{aligned}$$

where the last inequality holds by [10, Remark to Lemma 2.12] and by defining the error probability

$$\epsilon_n \triangleq \Pr\left[g^{(n)}\left(\phi^{(n)}(X^n), Y^n\right) \neq X^n\right], \tag{27}$$

which by assumption (21) satisfies

$$\overline{\lim_{n\to\infty}} \epsilon_n \leq \epsilon. \tag{28}$$

From (26) and because the error probability is bounded away from one, see (28), using (3b) we obtain:

$$\lim_{n\to\infty} \frac{1}{n} \log \Delta_n = 0. \tag{29}$$

Define the new random tuples $(\tilde{X}^n, \tilde{Y}^n)$ of joint pmf

$$P_{\tilde{X}^n\tilde{Y}^n}(x^n, y^n) = \frac{P_{X^nY^n}(x^n, y^n)}{\Delta_n} \cdot \mathbb{1}\{(x^n, y^n) \in \mathcal{D}_n\}, \tag{30}$$

and let $\tilde{\mathsf{M}} = \phi^{(n)}(\tilde{X}^n)$ and T be uniform over $\{1, \dots, n\}$ independent of $(\tilde{X}^n, \tilde{Y}^n, \tilde{\mathsf{M}})$.

Bounding the Rate: The rate is bounded as:

$$R \geq \frac{1}{n} H(\tilde{\mathsf{M}}) \geq \frac{1}{n} H(\tilde{\mathsf{M}}|\tilde{Y}^n) \overset{(a)}{=} \frac{1}{n} H(\tilde{X}^n|\tilde{Y}^n) \tag{31}$$
$$= H(X|Y) + o(1), \tag{32}$$

where Equality (a) holds because under our new measure $P_{\tilde{X}^n\tilde{Y}^n}$ reconstruction errors have zero probability and thus $\tilde{X}^n$ can be obtained as a function of $\tilde{\mathsf{M}}$ and $\tilde{Y}^n$ and (32) follows by Limits (10) in Lemma 1. Letting $n \to \infty$ establishes the desired result. ■

Above lossless source coding example well illustrates the idea of change of measure strong converse proofs and why we would like to restrict the new measure to the jointly typical set. With the new measure we transform the problem into a zero-error problem (because the set $\mathcal{D}_n$ only includes source sequences leading to perfect reconstructions), thus avoiding cumbersome error terms and immediately obtaining the equality in (31). Moreover, the conditional entropy of the new measure tends to the one of the original i.i.d. measure, because the former is restricted to the jointly typical set and compared to the original i.i.d. measure is boosted at most by the factor Δ_n^{-1}, which does not scale exponentially in the blocklength and only contributes to a $o(1)$ term, see (32).

Gu and Effros gave a related strong converse proof in [7]. Instead of the bounding steps (31), their proof bounds the size of the number of source sequences x^n for which (x^n, y^n) lies in $\mathcal{D}_n$ for a specific sequence y^n, which by the zero-error property of set $\mathcal{D}_n$ lower-bounds 2^{nR}.

If one was to adopt the Tyagi and Watanabe strong converse proof method to this setup, one would define $\mathcal{D}_n$ as the set of *all sequences* (x^n, y^n) satisfying $g^{(n)}(\phi^{(n)}(x^n), y^n) = x^n$, even if they are not jointly typical. The Tyagi-Watanabe converse proof then defines random sequences $(\tilde{X}^n, \tilde{Y}^n)$ according to the pmf (30), but now with the new definition of $\mathcal{D}_n$. The new message $\tilde{\mathsf{M}}$ is defined as in the presented proof and the same converse steps leading to (31) do apply. Since the sequences $(\tilde{X}^n, \tilde{Y}^n)$ do not necessarily lie in the jointly typical set, in the Tyagi and Watanabe approach one has to introduce a variational characterization of the right-hand side of (31) including the original product distribution $P_X^{\otimes n}$, single-letterize this variational characterization, and then show that the single-letter form is the variational characterization of the quantity in (32).

Remark 1: The error probability bound (28) only serves to prove (29). Since (29) remains valid when (28) is relaxed to

$$\lim_{n\to\infty} \frac{1}{n} \log(1 - \epsilon_n) = 0, \tag{33}$$

our proof technique can be used to show the so called *exponentially-strong converse*, which establishes that whenever $R < H(X|Y)$, then the probability of error behaves as $1 - e^{-n\theta}$ for some positive $\theta > 0$.

B. ϵ-Dependent Converse under an Expected-Rate Constraint

Consider the expected rate-constraint (20).

Definition 2: Given $\epsilon \in [0, 1)$. An *expected-rate* $R_{\text{avg}} > 0$ is ϵ-achievable if there exists a sequences $\{(\phi^{(n)}, g^{(n)})\}_{n=1}^{\infty}$ satisfying (20) and (21).

Theorem 2: For any $\epsilon \in [0, 1)$, any expected rate $R_{\text{avg}} < (1 - \epsilon)H(X|Y)$ is not ϵ-achievable.

The theorem states that if one only allows for an expected rate $R_{\text{avg}} = (1 - \epsilon)H(X|Y)$, then the probability of *correct* reconstruction is at most $(1 - \epsilon)$ asymptotically as $n \to \infty$.

Proof: Fix a sequence $\{(\phi^{(n)}, g^{(n)})\}_{n=1}^{\infty}$ satisfying (20) and (21).

Change of Measure: Same as in Subsection III-A.

Bounding the Rate: We start by bounding the rate as:

$$R_{\text{avg}} \geq \frac{1}{n} \mathbb{E}[\text{len}(\mathsf{M})] \tag{34}$$
$$= \frac{1}{n} \mathbb{E}[\text{len}(\mathsf{M})|(X^n, Y^n) \in \mathcal{D}_n]\Delta_n + \frac{1}{n} \underbrace{\mathbb{E}[\text{len}(\mathsf{M})|(X^n, Y^n) \notin \mathcal{D}_n]}_{\geq 0}(1 - \Delta_n) \tag{35}$$
$$\geq \frac{1}{n} \mathbb{E}[\text{len}(\tilde{\mathsf{M}})]\Delta_n \tag{36}$$
$$\geq \frac{1}{n} H(\tilde{\mathsf{M}}|\text{len}(\tilde{\mathsf{M}}))\Delta_n \tag{37}$$
$$= \frac{1}{n} \left(H(\tilde{\mathsf{M}}) - H(\text{len}(\tilde{\mathsf{M}})) \right) \Delta_n, \tag{38}$$

where the last step holds by the chain rule and because $H(\tilde{\mathsf{M}}) = H(\tilde{\mathsf{M}}, \text{len}(\tilde{\mathsf{M}}))$.

We continue to show that the entropy term $H(\text{len}(\tilde{\mathsf{M}}))$ grows sublinearly in the blocklength n. This holds because $\text{len}(\tilde{\mathsf{M}})$ is a non-negative integer-valued random variable with first moment bounded by $\frac{nR_{\text{avg}}}{\Delta_n}$, see (36), which implies that its entropy $H(\text{len}(\tilde{\mathsf{M}}))$ is bounded by the entropy of the geometric distribution with mean $\frac{nR_{\text{avg}}}{\Delta_n}$. I.e.,

$$\frac{1}{n} H(\text{len}(\tilde{\mathsf{M}})) \leq \frac{R_{\text{avg}}}{\Delta_n} h_b \left(\frac{\Delta_n}{nR_{\text{avg}}} \right) = o(1). \tag{39}$$

Combined with (38), we thus obtain:

$$R_{\text{avg}} \geq \frac{1}{n} H(\tilde{\mathsf{M}}) \cdot \Delta_n + o(1), \tag{40}$$

and following the same steps as in (31), we can further conclude:

$$R_{\text{avg}} \geq \Delta_n \cdot H(X|Y) + o(1). \tag{41}$$

The strong converse is then concluded by letting $n \to \infty$ and recalling that $\underline{\lim}_{n\to\infty} \Delta_n \geq (1 - \epsilon)$, see (26) and (28). ■

IV. Lossy Compression with Side-Information

Reconsider the compression problem in Figure 1, but where now the reconstructed sequence $\hat{X}^n$ only has to match the source X^n approximately according to a distortion function and maximum allowed distortion D. The error event is thus:

$$\frac{1}{n} \sum_{i=1}^{n} d(X_i, \hat{X}_i) \geq D, \tag{42}$$

for a given non-negative and bounded per-symbol distortion function $d(\cdot, \cdot)$. Lossy source coding with receiver side-information was first considered by Wyner and Ziv [16]. The

(exponentially-) strong converse was proved by Oohama [17] and its counterpart without receiver side-information by Körner [18].

In the following, we prove converse results using our change of measure technique. Two additional ingredients will be required compared to our proofs in the lossless case:

- Obtaining the single-letter rate-constraint (45) requires more complicated steps, e.g., Csiszár 'sum-identity.
- An auxiliary random variable U has to be identified and the Markov chain $U \to X \to Y$ needs to be proved. The Markov chain will be proved in an asymptotic sense for increasing blocklengths $n \to \infty$.

A. Strong Converse Under a Maximum-Rate Constraint

Definition 3: Given a distortion function $d(\cdot,\cdot)$. A *maximum-rate* $R > 0$ is said (ϵ, D)-achievable if there exists a sequence $\{(\phi^{(n)}, g^{(n)})\}_{n=1}^{\infty}$ satisfying (19) and

$$\overline{\lim_{n\to\infty}} \Pr\left[\frac{1}{n}\sum_{i=1}^{n} d(X_i, \hat{X}_i) > D\right] \leq \epsilon. \tag{43}$$

Theorem 3 ([17]): Given $\epsilon \in [0,1)$ and maximum distortion $D \geq 0$. Maximum rate $R > 0$ is not (ϵ, D)-achievable, if

$$R < \mathsf{R}_{WZ}(D), \tag{44}$$

where

$$\mathsf{R}_{WZ}(D) \triangleq \max_{U} I(X; U|Y) \tag{45}$$

for auxiliary random variables U, Z satisfying the Markov chains $U \to X \to Y$ and $X \to (U, Y) \to Z$ and the distortion constraint

$$D \geq \mathbb{E}[d(X, Z)]. \tag{46}$$

Proof of Theorem 3: Fix a sequence $\{(\phi^{(n)}, g^{(n)})\}_{n=1}^{\infty}$ satisfying the maximum rate-constraint (19) and excess distortion criterion (43).

Change of Measure: Define the subset

$$\mathcal{D}_n := \left\{(x^n, y^n) \in \mathcal{T}_{\mu_n}^{(n)}(P_{XY}) \colon \frac{1}{n}\sum_{i=1}^{n} d(x_i, \hat{x}_i) < D\right\}, \tag{47}$$

where $\hat{x}^n$ denotes the reconstruction sequence $\hat{x}^n = g^{(n)}(\phi^{(n)}(x^n), y^n)$. Denote the probability of set $\mathcal{D}_n$ by Δ_n. Similarly to Subsection III-A and because the probability of excess distortion is bounded away from 1, see (43), we have:

$$\underline{\lim_{n\to\infty}} \Delta_n \geq 1 - \epsilon \tag{48a}$$

$$\lim_{n\to\infty} \frac{1}{n} \log \Delta_n = 0. \tag{48b}$$

Define $(\tilde{X}^n, \tilde{Y}^n, \tilde{\mathsf{M}}, T)$ as in Subsection III-A but with the new definition of $\mathcal{D}_n$ in (47). Thus, $(\tilde{X}^n, \tilde{Y}^n)$ have joint pmf

$$P_{\tilde{X}^n\tilde{Y}^n}(x^n, y^n) = \frac{P_{XY}^{\otimes n}(x^n, y^n)}{\Delta_n} \cdot \mathbb{1}\{(x^n, y^n) \in \mathcal{D}_n\}, \tag{49}$$

T is uniform over $\{1, \ldots, n\}$ independent of $(\tilde{X}^n, \tilde{Y}^n)$, and

$$\tilde{\mathsf{M}} \triangleq \phi^{(n)}(\tilde{X}^n), \tag{50}$$

$$\tilde{Z}^n \triangleq g^{(n)}(\tilde{Y}^n, \tilde{\mathsf{M}}), \tag{51}$$

$$\tilde{U} \triangleq (\tilde{X}^{T-1}, \tilde{Y}_{T+1}^n, \tilde{\mathsf{M}}, T). \tag{52}$$

Bounding the Rate: We notice that:

$$\begin{aligned}
R &\geq \frac{1}{n} H(\tilde{\mathsf{M}}) \geq \frac{1}{n} H(\tilde{\mathsf{M}}|\tilde{Y}^n) && (53)\\
&\overset{(a)}{=} \frac{1}{n} I(\tilde{\mathsf{M}}; \tilde{X}^n|\tilde{Y}^n) && (54)\\
&= \frac{1}{n}\left[H(\tilde{X}^n|\tilde{Y}^n) - H(\tilde{X}^n|\tilde{Y}^n, \tilde{\mathsf{M}})\right] && (55)\\
&\overset{(b)}{=} H(\tilde{X}_T|\tilde{Y}_T) - H(\tilde{X}_T|\tilde{Y}_T, \tilde{U}) + o(1) && (56)\\
&= I(\tilde{U}; \tilde{X}_T|\tilde{Y}_T) + o(1), && (57)
\end{aligned}$$

where (a) holds because $\tilde{\mathsf{M}}$ is a function of $\tilde{X}^n$; and (b) holds by Lemmas 1 and 2 and the definition of $\tilde{U}$ in (52).

Approximate Markov Conditions: Since $\tilde{\mathsf{M}}$ is a function of $\tilde{X}^n$:

$$\begin{aligned}
0 &= \frac{1}{n} I(\tilde{\mathsf{M}}; \tilde{Y}^n|\tilde{X}^n) && (58)\\
&= \frac{1}{n}\left[H(\tilde{Y}^n|\tilde{X}^n) - H(\tilde{Y}^n|\tilde{X}^n, \tilde{\mathsf{M}})\right] && (59)\\
&\overset{(c)}{\geq} H(\tilde{Y}_T|\tilde{X}_T) + o(1) - H(\tilde{Y}_T|\tilde{X}_T, \tilde{U}) && (60)\\
&= I(\tilde{U}; \tilde{Y}_T|\tilde{X}_T) + o(1) && (61)\\
&\geq o(1), && (62)
\end{aligned}$$

where (c) holds by Lemmas 1 and 2 and the definition of $\tilde{U}$ in (52). Above sequence of inequalities implies that

$$I(\tilde{U}; \tilde{Y}_T|\tilde{X}_T) = o(1). \tag{63}$$

In a similar way, since $\tilde{Z}^n$ is a function of $(\tilde{Y}^n, \tilde{\mathsf{M}})$:

$$\begin{aligned}
0 &= \frac{1}{n} I(\tilde{Z}^n; \tilde{X}^n|\tilde{Y}^n, \tilde{\mathsf{M}}) && (64)\\
&= \frac{1}{n}\left[H(\tilde{X}^n|\tilde{Y}^n, \tilde{\mathsf{M}}) - H(\tilde{X}^n|\tilde{Y}^n, \tilde{\mathsf{M}}, \tilde{Z}^n)\right] && (65)\\
&\overset{(d)}{\geq} H(\tilde{X}_T|\tilde{Y}_T, \tilde{U}) + o(1) && (66)\\
&\quad - \frac{1}{n}\sum_{i=1}^{n} H(\tilde{X}_i|\tilde{X}^{i-1}, \tilde{Y}_i, \tilde{Y}_{i+1}^n, \tilde{\mathsf{M}}, \tilde{Z}_i) && (67)\\
&\geq I(\tilde{Z}_T; \tilde{X}_T|\tilde{Y}_T, \tilde{U}) + o(1), && (68)
\end{aligned}$$

where (d) holds by Lemma 2 and because conditioning cannot increase entropy. Thus,

$$I(\tilde{Z}_T; \tilde{X}_T|\tilde{Y}_T, \tilde{U}) = o(1). \tag{69}$$

In words, Equations (63) and (69) imply that as $n \to \infty$ the two random variables $\tilde{Y}_T$ and $\tilde{U}$ become independent given $\tilde{X}_T$ and the random variables $\tilde{Z}_T$ and $\tilde{X}_T$ become independent given $(\tilde{Y}_T, \tilde{U})$.

Bounding the Distortion: Since subset $\mathcal{D}_n$ only contains sequences (x^n, y^n) for which $\frac{1}{n}\sum_{i=1}^{n} d(x_i, z_i) \leq D$, we have:

$$D \geq \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}\left[d\left(\tilde{X}_i, \tilde{Z}_i\right)\right] = \mathbb{E}\left[d\left(\tilde{X}_T, \tilde{Z}_T\right)\right], \tag{70}$$

where the equality holds by the definition of T.

Asymptotic Analysis: The theorem follows by combining (57), (63), (69), and (70) with (7) in Lemma 1 and letting the blocklength $n \to \infty$. More specificially, we can first conclude

by Carathéodory's theorem [19, Appendix C] that there exists an auxiliary random variable $\tilde{U}$ of bounded alphabet satisfying (57), (63), (69), and (70). We restrict to such auxiliary random variables and invoke the Bolzano-Weierstrass theorem to conclude the existence of a pmf P^*_{UXYZ} and an increasing subsequence of blocklengths $\{n_i\}_{i=1}^{\infty}$ so that

$$\lim_{n_i\to\infty} P_{\tilde{U}\tilde{X}_T\tilde{Y}_T\tilde{Z}_T} = P^*_{UXYZ}, \tag{71}$$

where we can conclude by (7) of Lemma 1 that

$$P^*_{\tilde{X}_T\tilde{Y}_T} = P_{XY}. \tag{72a}$$

We further deduce from (57), (63), and (69) that:

$$R \geq I_{P^*}(X;U|Y) \tag{72b}$$

and

$$0 = I_{P^*}(Y;U|X) \tag{72c}$$

$$0 = I_{P^*}(Z;X|Y,U) \tag{72d}$$

where the subscript P^* indicates that the mutual information quantities should be computed with respect to P^*.

Combined with (70), which implies

$$D \geq \mathbb{E}_{P^*}[d(X,Z)], \tag{73}$$

above (in)equalities (72) conclude the desired converse proof. ■

The rate-bounding steps (53)–(57) and the proof steps in the asymptotic Markov conditions (58)–(62) are identical to steps in [9]. But while here Steps (58)–(62) serve to prove asymptotic Markov chains, in [9] they are used to single-letterize the variational characterizations.

Our technique can also be used to show other strong converse results, for example for interactive source coding [1]. More strong converse results for network source coding problems can be found in [7]–[9], [20], [21].

B. ϵ-Dependent Converse under an Expected-Rate Constraint

Definition 4: Given $\epsilon \in [0,1)$ and $D \geq 0$. An *expected-rate* $R_{\text{avg}} > 0$ is said (ϵ, D)-achievable if there exists a sequence $\{(\phi^{(n)}, g^{(n)})\}_{n=1}^{\infty}$ satisfying (20) and (43).

Theorem 4 ([22]): For any $\epsilon \in [0,1)$ and $D > 0$. An expected-rate R_{avg} is not (ϵ, D)-achievable if

$$R_{\text{avg}} < (1-\epsilon)\mathsf{R}_{\text{WZ}}(D). \tag{74}$$

Proof: Same proof steps as in the previous Section IV-A, but replace the rate-bounding paragraph by the following steps.

<u>*Bounding the Rate:*</u> Apply Steps (34)–(39) to obtain

$$R_{\text{avg}} \geq \frac{1}{n} H(\tilde{\mathsf{M}}) \cdot \Delta_n + o(1), \tag{75}$$

then bound $\frac{1}{n}H(\tilde{\mathsf{M}})$ as in (53)–(57) to obtain:

$$R_{\text{avg}} \geq I(\tilde{U}; \tilde{X}_T|\tilde{Y}_T) \cdot \Delta_n + o(1). \tag{76}$$

■

Remark 2 (General Expectation Constraints): A main feature of the rate-bounding steps (34)–(39) is to write the expectation into conditional expectations over the set $\mathcal{D}_n$ and its complement $\mathcal{D}_n^c$ and then bound the expectation over $\mathcal{D}_n^c$ using a trivial bound (0 in case of the expected length). The rest of the proof then focuses on bounding the expectation over the set $\mathcal{D}_n^c$ where one can exploit the specific structure (the zero-error and typicality assumptions) in this set.

Above idea can also be used to bound other expectation constraints, e.g., the equivocation constraint $H(X^n|\mathsf{M})$.

C. Two Distortions under Maximum-Rate Constraints

Consider now the setup of Section IV-A but where in Definition 6 we have pairs of distortions (D_1, D_2) and distortion functions (d_1, d_2), and where the reconstructed sequence $\hat{X}^n$ simultaneously has to satisfy two excess distortion constraints:

$$\overline{\lim_{n\to\infty}} \Pr\left[\frac{1}{n}\sum_{i=1}^{n} d_1(X_i, \hat{X}_i) > D_1\right] \leq \epsilon_1 \tag{77a}$$

$$\overline{\lim_{n\to\infty}} \Pr\left[\frac{1}{n}\sum_{i=1}^{n} d_2(X_i, \hat{X}_i) > D_2\right] \leq \epsilon_2. \tag{77b}$$

Under a maximum-rate constraint we have:

Theorem 5 ($\epsilon_1 + \epsilon_2 < 1$): Given maximum distortions $D_1, D_2 \geq 0$ and $\epsilon_1, \epsilon_2 \in [0,1)$ satisfying

$$\epsilon_1 + \epsilon_2 < 1. \tag{78}$$

A maximum rate R is not $(\epsilon_1, \epsilon_2, D_1, D_2)$-achievable if

$$R < \mathsf{R}_{\text{WZ-2}}(D_1, D_2), \tag{79}$$

where

$$\mathsf{R}_{\text{WZ-2}}(D_1, D_2) \triangleq \max_U I(X;U|Y) \tag{80}$$

for auxiliary random variables U, Z satisfying the Markov chain $U \to X \to Y$ and $X \to (U,Y) \to Z$ and the two reconstruction constraints

$$\mathbb{E}[d_1(X,Z)] \leq D_1 \tag{81}$$

$$\mathbb{E}[d_2(X,Z)] \leq D_2. \tag{82}$$

In contrast to the single-distortion rate distortion function $\mathsf{R}_{\text{WZ}}(\cdot)$ here the reconstruction random variable Z has to satisfy two distortion constraints. To be able to find a suitable Z an enhanced auxiliary U_2 might be required in the two-distortions case compared to the auxiliary U that maximizes $\mathsf{R}_{\text{WZ}}(\cdot)$. In other words, all auxiliaries U_2 allowing for a Z satisfying (81) and (82) satisfy $I(U_2;X|Y) \geq I(U;X|Y)$ implying the inequality $\mathsf{R}_{\text{WZ}-2}(\cdot) \geq \mathsf{R}_{\text{WZ}}(\cdot)$ and in certain cases the inequality is strict.

Proof: Follow the proof steps in Section IV-A, but subject to the following two modifications.

<u>*Modifications in the Change of Measure:*</u> Use both distortion constraints in the definition of $\mathcal{D}_n$:

$$\mathcal{D}_n \triangleq \left\{(x^n, y^n) \in \mathcal{T}_{\mu_n}^{(n)}(P_{XY}) \colon \frac{1}{n}\sum_{i=1}^{n} d_1(x_i, \hat{x}_i) < D_1, \frac{1}{n}\sum_{i=1}^{n} d_2(x_i, \hat{x}_i) < D_2\right\}. \tag{83}$$

Notice that by the union bound and (77), we have:

$$\overline{\lim_{n\to\infty}} \Pr\left[\frac{1}{n}\sum_{i=1}^{n} d_1(X_i, \hat{X}_i) < D_1 \text{ and } \frac{1}{n}\sum_{i=1}^{n} d_2(X_i, \hat{X}_i) < D_2\right]$$

$$\geq 1 - \epsilon_1 - \epsilon_2 > 0, \tag{84}$$

where the strict inequality in (84) holds because $\epsilon_1 + \epsilon_2 < 1$.

The newly defined set $\mathcal{D}_n$ satisfies all the requirements of the proof steps in Section IV-A. In particular, (84) ensures Limit (48b).

Modifications when Bounding the Distortions: Apply (70) for both (d_1, D_1) and (d_2, D_2). ■

Remark 3 ($\epsilon_1 + \epsilon_2 \geq 1$): Theorem 5 cannot be strengthened to arbitrary $\epsilon_1, \epsilon_2 \in [0, 1)$. In fact, the theorem is generally not valid when $\epsilon_1 + \epsilon_2 \geq 1$. In this latter case, $(1 - \epsilon_1) + (1 - \epsilon_2) < 1$ and the encoder can choose with probability $1 - \epsilon_1$ a scheme (using an auxiliary U_1) that is tailored to achieve a small distortion under $d_1(\cdot, \cdot)$ and with probability ϵ_2 a scheme (using a different auxiliary U_2) that is tailored to achieve a small distortion under $d_2(\cdot, \cdot)$. To indicate the choice of the scheme to the decoder, the encoder can simply add a single flagbit to its message (e.g., at the beginning).

The described scheme is optimal whenever $\epsilon_1 + \epsilon_2 \geq 1$ and a matching converse can be proved by applying the change-of-measure proof in Section IV-A twice, once for distortion function d_1 and maximum distortion D_1 and the second time for d_2 and D_2. So, for $\epsilon_1 + \epsilon_2 \geq 1$, Bound (79) in Theorem 5 needs to be changed to

$$R < \max\{\mathsf{R}_{\text{WZ}}(D_1),\ \mathsf{R}_{\text{WZ}}(D_2)\}, \tag{85}$$

where the two terms $\mathsf{R}_{\text{WZ}}(D_1)$ and $\mathsf{R}_{\text{WZ}}(D_2)$ are with respect to the two distortion functions d_1 and d_2 respectively.

D. Two Distortions under Expected-Rate Constraints

Reconsider the setup of Section IV-B with pairs of distortions (D_1, D_2) and distortion functions (d_1, d_2), but now under an expected-rate constraint.

Theorem 6: Given $1 > \epsilon_1 \geq \epsilon_2 \geq 0$ and distortion functions d_1, d_2. For $D_1, D_2 \geq 0$ expected-rate R_{avg} is not $(\epsilon_1, \epsilon_2, D_1, D_2)$-achievable if

$$R < (1 - \epsilon_1)\mathsf{R}_{\text{WZ-2}}(D_1, D_2) + (\epsilon_1 - \epsilon_2)\mathsf{R}_{\text{WZ}}(D_2), \tag{86}$$

where $\mathsf{R}_{\text{WZ}}(D_2)$ is with respect to d_2.

Remark 4: In general, $\mathsf{R}_{\text{WZ}}(D_2) \leq \mathsf{R}_{\text{WZ-2}}(D_1, D_2)$, and the inequality is strict in interesting special cases. In these cases the minimum compression rate $R < (1 - \epsilon_2)\mathsf{R}_{\text{WZ-2}}(D_1, D_2)$, and it is thus possible to reduce the compression rate even if only one of the two distortion constraints can be violated with a higher probability.

When $\epsilon_1 = \epsilon_2$ then the minimum compression rate is given by $(1 - \epsilon_2)^{-1}\mathsf{R}_{\text{WZ-2}}(D_1, D_2)$ and thus imposing an expected-rate constraint instead of a maximum-rate constraint allows to reduce the rate the compression rate by a factor of $(1 - \epsilon_2)$.

Remark 5: The rate in the theorem is achieved by a scheme that with probability $(1 - \epsilon_1)$ applies an optimal rate-distortion code for the two distortions D_1 and D_2 and with probability $\epsilon_1 - \epsilon_2$ applies a rate-distortion code for distortion D_2 only.

Proof of Theorem 6: Our proof combines two instances of the proof in Subsection IV-A (which holds under a single distortion constraint) with an instance of the proof in Subsection IV-C (which holds under two distortion constraints), followed by some optimization-theoretic considerations.

Fix a sequence $\{(\phi^{(n)}, g^{(n)})\}_{n=1}^{\infty}$ satisfying (19) and (77).

Parallel Changes of Measure: We employ three changes of measure in parallel, on three disjoint subsets $\mathcal{D}_{12}, \mathcal{D}_1, \mathcal{D}_2$ of the typical set $\mathcal{T}^{(n)}(P_{XY})$.

Apply the change of measure in Subsection IV-C, i.e., based on the set

$$\mathcal{D}_{12,n} \triangleq \left\{(x^n, y^n) \in \mathcal{T}_{\mu_n}^{(n)}(P_{XY})\colon \frac{1}{n}\sum_{i=1}^{n} d_1(x_i, \hat{x}_i) < D_1,\ \frac{1}{n}\sum_{i=1}^{n} d_2(x_i, \hat{x}_i) < D_2\right\}, \tag{87}$$

and denote the induced random variables (see Subsection IV-C) with an additional subscript $_{12}$. I.e., denote them $\tilde{X}_{12}^n, \tilde{Y}_{12}^n, \tilde{\mathsf{M}}_{12}, \tilde{Z}_{12}^n, \tilde{U}_{12}$.

Apply the change of measure in Subsection IV-A based on the set

$$\mathcal{D}_{1,n} \triangleq \left\{(x^n, y^n) \in \mathcal{T}_{\mu_n}^{(n)}(P_{XY})\colon \frac{1}{n}\sum_{i=1}^{n} d_1(x_i, \hat{x}_i) < D_1,\ \frac{1}{n}\sum_{i=1}^{n} d_2(x_i, \hat{x}_i) \geq D_2\right\}, \tag{88}$$

and denote the induced random variables (see Subsection IV-A) with an additional subscript $_1$. I.e., $\tilde{X}_1^n, \tilde{Y}_1^n, \tilde{\mathsf{M}}_1, \tilde{Z}_1^n, \tilde{U}_1$.

Finally, apply a change of measure in Subsection IV-A based on the set

$$\mathcal{D}_{2,n} \triangleq \left\{(x^n, y^n) \in \mathcal{T}_{\mu_n}^{(n)}(P_{XY})\colon \frac{1}{n}\sum_{i=1}^{n} d_1(x_i, \hat{x}_i) \geq D_1,\ \frac{1}{n}\sum_{i=1}^{n} d_2(x_i, \hat{x}_i) < D_2\right\}, \tag{89}$$

and denote the induced random variables (see Subsection IV-A) with an additional subscript $_2$. . I.e., $\tilde{X}_2^n, \tilde{Y}_2^n, \tilde{\mathsf{M}}_2, \tilde{Z}_2^n, \tilde{U}_2$.

Denote the probabilities of the three sets by $\Delta_{12,n}, \Delta_{1,n}$, and $\Delta_{2,n}$, which by (77) satisfy the following limiting behaviours

$$\varliminf_{n\to\infty} (\Delta_{1,n} + \Delta_{12,n}) \geq 1 - \epsilon_1 \tag{90}$$

$$\varliminf_{n\to\infty} (\Delta_{2,n} + \Delta_{12,n}) \geq 1 - \epsilon_2 \tag{91}$$

$$\varliminf_{n\to\infty} \Delta_{12,n} \geq 1 - \epsilon_1 - \epsilon_2. \tag{92}$$

Asymptotic Markov Chains: For each of the defined set of auxiliaries (with subscripts $_{12}$, $_1$, $_2$) we can prove the asymptotic Markov chains as in Section IV-A.

Bounding the Distortions: Similarly to the proof in Subsection IV-C, for the auxiliaries with subscript $_{12}$ we can prove the two distortion constraints

$$D_k \geq \mathbb{E}\left[d_k\left(\tilde{X}_{12,T}, \tilde{Z}_{12,T}\right)\right], \quad k \in \{1, 2\}. \tag{93}$$

For the auxiliaries with subscript $_1$ we can show a single distortion constraint as in Subsection IV-A:

$$D_1 \geq \mathbb{E}\left[d_1\left(\tilde{X}_{1,T}, \tilde{Z}_{1,T}\right)\right], \tag{94}$$

and similarly, for the auxiliaries with subscript $_2$:

$$D_2 \geq \mathbb{E}\left[d_2\left(\tilde{X}_{2,T}, \tilde{Z}_{2,T}\right)\right]. \tag{95}$$

Bounding the Rate: Since the subsets $\mathcal{D}_{12}, \mathcal{D}_1, \mathcal{D}_2$ are disjoint and because a length is non-negative, by the total law of expectation we have:

$$R_{\text{avg}} \geq \frac{1}{n}\mathbb{E}[\text{len}(\mathsf{M})] \tag{96}$$

$$\geq \frac{1}{n}\mathbb{E}[\text{len}(\tilde{\mathsf{M}}_{12})]\Delta_{12,n} + \frac{1}{n}\mathbb{E}[\text{len}(\tilde{\mathsf{M}}_1)]\Delta_{1,n} + \frac{1}{n}\mathbb{E}[\text{len}(\tilde{\mathsf{M}}_2)]\Delta_{2,n}. \tag{97}$$

Each of the expectations $\mathbb{E}[\text{len}(\tilde{\mathsf{M}}_{12})]$, $\mathbb{E}[\text{len}(\tilde{\mathsf{M}}_1)]$, and $\mathbb{E}[\text{len}(\tilde{\mathsf{M}}_2)]$ can then be bounded as in (34)–(39) and further as in (53)–(57) to obtain

$$R_{\text{avg}} \geq I(\tilde{U}_{12}; \tilde{X}_{12,T}|\tilde{Y}_{12,T})\Delta_{12,n} + I(\tilde{U}_1; \tilde{X}_{1,T}|\tilde{Y}_{1,T})\Delta_{1,n} + I(\tilde{U}_2; \tilde{X}_{2,T}|\tilde{Y}_{2,T})\Delta_{2,n} + o(1). \tag{98}$$

Asymptotic Analysis: Letting $n \to \infty$, considering an appropriate subsequence, and invoking Carathéodory's theorem, one can show similarly as in Subsection IV-A that

$$R_{\text{avg}} \geq I(U_{12}; X|Y)\Delta_{12} + I(U_1; X|Y)\Delta_1 + I(U_2; X|Y)\Delta_2 \tag{99}$$

(where $(X, Y) \sim P_{XY}$) for auxiliaries $U_1, U_2, U_{12}, Z_1, Z_2, Z_{12}$ satisfying the Markov chains $U_k \to X \to Y$ and $X \to (U_k, Y) \to Z_k$, for $k \in \{1, 2, 12\}$, and the distortion constraints

$$D_1 \geq \mathbb{E}\left[d_1\left(X, \tilde{Z}_k\right)\right], \quad k \in \{1, 12\} \tag{100}$$

and

$$D_2 \geq \mathbb{E}\left[d_2\left(X, \tilde{Z}_k\right)\right], \quad k \in \{2, 12\}, \tag{101}$$

and probabilities $\Delta_1, \Delta_2, \Delta_{12} \geq 0$ satisfying

$$\Delta_1 + \Delta_{12} \geq 1 - \epsilon_1 \tag{102a}$$

$$\Delta_2 + \Delta_{12} \geq 1 - \epsilon_1 \tag{102b}$$

$$\Delta_{12} \geq 1 - \epsilon_1 - \epsilon_2. \tag{102c}$$

Simplifying the Solution: Using Bounds (102), rate-constraint (99) can be relaxed to

$$R_{\text{avg}} \geq I(U_{12}; X|Y)\Delta_{12} + I(U_1; X|Y)[1 - \epsilon_1 - \Delta_{12}]^+ + I(U_2; X|Y)[1 - \epsilon_2 - \Delta_{12}]^+, \tag{103}$$

subject to $\Delta_{12} \geq 1 - \epsilon_1 - \epsilon_2$.

It can be shown that without loss of optimality one can restrict to auxiliaries so that

$$\max\{I(U_1; X|Y),\ I(U_2; X|Y)\} \leq I(U_{12}; X|Y) \leq I(U_1; X|Y) + I(U_2; X|Y). \tag{104}$$

Under this assumption, Constraint (103) is loosest for $\Delta_{12} = 1 - \max\{\epsilon_1, \epsilon_2\}$. Plugging this choice into (103) results in the desired expression of the theorem. ∎

V. Binary Hypothesis Testing

We focus on the famous Stein binary hypothesis testing problem where the false-alarm error probability has to stay below a given threshold asymptotically for infinite blocklengths while the miss-detection error probability has to decay exponentially fast to 0 with largest possible exponent.

A. Centralized Testing

Consider a single decision center observing a sequence Y^n that is distributed according to one of two distributions depending on a binary hypothesis $\mathcal{H} \in \{0, 1\}$:

$$\text{if } \mathcal{H} = 0\colon \quad Y^n \text{ i.i.d. } \sim P_Y; \tag{105a}$$

$$\text{if } \mathcal{H} = 1\colon \quad Y^n \text{ i.i.d. } \sim Q_Y, \tag{105b}$$

for given pmfs P_Y and Q_Y over the same alphabet $\mathcal{Y}$.

The decision center produces a guess

$$\hat{\mathcal{H}} = g^{(n)}(Y^n) \in \{0, 1\}. \tag{106}$$

We have the following achievability definition and result.

Definition 5: Given a permissible false-alarm error probability $\epsilon \in [0, 1)$. A miss-detection error exponent $\theta > 0$ is called ϵ-achievable if there exists a sequence $\{g^{(n)}\}_{n=1}^{\infty}$ satisfying

$$\overline{\lim_{n\to\infty}} \Pr\left[\hat{\mathcal{H}} = 1|\mathcal{H} = 0\right] \leq \epsilon \tag{107a}$$

$$\underline{\lim_{n\to\infty}} -\frac{1}{n}\log \Pr\left[\hat{\mathcal{H}} = 0|\mathcal{H} = 1\right] \geq \theta. \tag{107b}$$

The following is a well-known result.

Theorem 7 (The converse to Stein's Lemma): All miss-detection error exponents

$$\theta \geq D(P_Y \| Q_Y) \tag{108}$$

are not ϵ-achievable for any $\epsilon \in [0, 1)$.

Proof: Fix a sequence $\{g^{(n)}\}_{n\geq 1}$ satisfying (107).

Change of Measure: Define the subset:

$$\mathcal{D}_n \triangleq \left\{y^n \in \mathcal{T}_{\mu_n}^{(n)}(P_Y)\colon g^{(n)}(y^n) = 0\right\} \tag{109}$$

and its probability under $\mathcal{H} = 0$:

$$\Delta_n \triangleq \Pr[Y^n \in \mathcal{D}_n|\mathcal{H} = 0]. \tag{110}$$

Since the probability of false-alarm

$$\alpha_n \triangleq \Pr\left[\hat{\mathcal{H}} = 1|\mathcal{H} = 0\right] \tag{111}$$

satisfies $\overline{\lim}_{n\to\infty} \alpha_n \leq \epsilon < 1$, we have similarly to the previous sections, e.g., see (24)–(29):

$$\lim_{n\to\infty} \frac{1}{n}\log \Delta_n = 0. \tag{112}$$

Define the new random tuple $\tilde{Y}^n$ to have pmf

$$P_{\tilde{Y}^n}(y^n) = \frac{P_Y^{\otimes n}(y^n)}{\Delta_n} \cdot \mathbb{1}\{y^n \in \mathcal{D}_n\}. \tag{113}$$

Bounding the Miss-Detection Error Exponent: Since the set $\mathcal{D}_n$ only contains sequences y^n resulting in $\hat{\mathcal{H}} = 0$, for the new measure we have $P_{\tilde{Y}^n}(\hat{\mathcal{H}} = 0) = 1$. Therefore:

$$D(P_{\tilde{Y}^n}(\hat{\mathcal{H}})\|Q_Y^{\otimes n}(\hat{\mathcal{H}})) = \log\frac{1}{Q_Y^{\otimes n}(\hat{\mathcal{H}} = 0)} = \log\frac{1}{\beta_n}, \tag{114}$$

where $\beta_n \triangleq \Pr[\hat{\mathcal{H}} = 0|\mathcal{H} = 1]$. Thus by the data-processing inequality for Kullback-Leibler divergence:

$$-\frac{1}{n}\log\beta_n = \frac{1}{n}D\left(P_{\tilde{Y}^n}(\hat{\mathcal{H}})\|Q_Y^{\otimes n}(\hat{\mathcal{H}})\right) \tag{115}$$

$$\leq \frac{1}{n}D\left(P_{\tilde{Y}^n}\|Q_Y^{\otimes n}\right) \tag{116}$$

$$\stackrel{(e)}{=} D(P_Y \| Q_Y) + o(1), \tag{117}$$

where (e) holds by Lemma 1. Combined with (112) this establishes the desired result by taking the limit $n \to \infty$. ■

B. Distributed Testing Against Independence

Consider the distributed hypothesis testing problem where a sensor observes a sequence X^n and communicates with a decision center that observes the sequence Y^n. The distribution of the observations (X^n, Y^n) depends on a binary hypothesis $\mathcal{H} \in \{0,1\}$:

$$\text{if } \mathcal{H} = 0: \quad (X^n, Y^n) \text{ i.i.d. } \sim P_{XY}; \tag{118a}$$

$$\text{if } \mathcal{H} = 1: \quad (X^n, Y^n) \text{ i.i.d. } \sim P_X P_Y, \tag{118b}$$

for a given pmf P_{XY} over the product alphabet $\mathcal{X} \times \mathcal{Y}$ and P_X and P_Y denoting the corresponding marginal pmfs.

The sensor conveys a message $\mathsf{M} = \phi^{(n)}(X^n) \in \{0,1\}^\star$ to the decision center, where the message is of maximum rate R:

$$\text{len}(\mathsf{M}) \leq nR. \tag{119}$$

Based on the received message M and its own observation Y^n, the decision center produces a guess of the hypothesis:

$$\hat{\mathcal{H}} = g^{(n)}(\mathsf{M}, Y^n) \in \{0,1\}. \tag{120}$$

Definition 6: Given $\epsilon \in [0,1)$ and a maximum rate $R > 0$. A miss-detection error exponent $\theta > 0$ is called (ϵ, R)-achievable if there exists a sequence $\{(f^{(n)}, g^{(n)})\}_{n=1}^\infty$ satisfying

$$\overline{\lim_{n\to\infty}} \Pr\left[\hat{\mathcal{H}} = 1 | \mathcal{H} = 0\right] \leq \epsilon \tag{121a}$$

$$\underline{\lim_{n\to\infty}} -\frac{1}{n} \log \Pr\left[\hat{\mathcal{H}} = 0 | \mathcal{H} = 1\right] \geq \theta. \tag{121b}$$

Theorem 8 ([23]): Given $\epsilon \in [0,1)$ and maximum rate $R > 0$. A miss-detection error exponent θ is not (ϵ, R)-achievable if satisfies

$$\theta > \max_{\substack{P_{V|X}: \\ I(V;X) \leq R}} I(V;Y), \tag{122}$$

where the mutual informations are calculated with respect to the joint pmf $P_{V|X} P_{XY}$.

Proof: Fix a sequence of encoding and decision functions $\{(\phi^{(n)}, g^{(n)})\}_{n \geq 1}$ satisfying (121).

Change of Measure: Define the set

$$\mathcal{D}_n \triangleq \{(x^n, y^n) \in \mathcal{T}_{\mu_n}^{(n)}(P_{XY}) \colon g^{(n)}(f^{(n)}(x^n), y^n) = 0\} \tag{123}$$

and denote its probability under $\mathcal{H} = 0$ by Δ_n. Since the probability of false-alarm

$$\alpha_n = \Pr\left[\hat{\mathcal{H}} = 1 | \mathcal{H} = 0\right] \tag{124}$$

satisfies $\overline{\lim}_{n\to\infty} \alpha_n \leq \epsilon < 1$, by steps similarly to (24)–(29), we conclude that

$$\lim_{n\to\infty} \frac{1}{n} \log \Delta_n = 0. \tag{125}$$

Let $(\tilde{X}^n, \tilde{Y}^n)$ be a pair of random tuples of joint pmf

$$P_{\tilde{X}^n \tilde{Y}^n}(x^n, y^n) = \frac{P_{XY}^{\otimes n}(x^n, y^n)}{\Delta_n} \cdot \mathbb{1}\{(x^n, y^n) \in \mathcal{D}_n\}, \tag{126}$$

and define $\tilde{M} = f^{(n)}(\tilde{X}^n)$ and T in the standard way and

$$V \triangleq (\tilde{X}^{T-1}, \tilde{Y}^{T-1}, \tilde{\mathsf{M}}, T). \tag{127}$$

Bounding the Miss-Detection Error Exponent:

Define

$$P_{\mathsf{M}} P_Y^{\otimes n}(\hat{\mathcal{H}} = 0) \triangleq \sum_{\substack{(\mathsf{m}, y^n): \\ g^{(n)}(\mathsf{m}, y^n) = 0}} P_{\mathsf{M}}(\mathsf{m}) P_Y^{\otimes n}(y^n) \tag{128}$$

and $P_{\mathsf{M}} P_{Y^n}(\hat{\mathcal{H}} = 1) = 1 - P_{\mathsf{M}} P_{Y^n}(\hat{\mathcal{H}} = 0)$.

Let the binary pmf $P_{\tilde{\mathsf{M}} \tilde{Y}^{\otimes n}}(\hat{\mathcal{H}})$ be defined in a similar way but with the product pmf $P_{\mathsf{M}}(\mathsf{m}) P_Y^{\otimes n}(y^n)$ replacing $P_{\tilde{\mathsf{M}} \tilde{Y}^{\otimes n}}$.

Notice that $P_{\tilde{\mathsf{M}} \tilde{Y}^n}(0) = 1$ and therefore:

$$\begin{aligned} D\left(P_{\tilde{\mathsf{M}} \tilde{Y}^n}(\hat{\mathcal{H}}) \| P_{\mathsf{M}} P_Y^{\otimes n}(\hat{\mathcal{H}})\right) &= \log \frac{1}{P_{\mathsf{M}} P_Y^{\otimes n}(\hat{\mathcal{H}} = 0)} \\ &= \log \frac{1}{\beta_n}, \end{aligned} \tag{129}$$

where

$$\beta_n \triangleq \Pr[\hat{\mathcal{H}} = 0 | \mathcal{H} = 1]. \tag{130}$$

Applying again the data-processing inequality for KL divergence, we obtain:

$$-\frac{1}{n} \log \beta_n = \frac{1}{n} D\left(P_{\tilde{M} \tilde{Y}^n}(\hat{\mathcal{H}}) \| P_M P_Y^{\otimes n}(\hat{\mathcal{H}})\right) \tag{131}$$

$$\leq \frac{1}{n} D\left(P_{\tilde{M} \tilde{Y}^n} \| P_M P_Y^{\otimes n}\right). \tag{132}$$

We next notice that

$$\begin{aligned} &\frac{1}{n} D\left(P_{\tilde{M} \tilde{Y}^n} \| P_M P_Y^{\otimes n}\right) \\ &\stackrel{(f)}{\leq} \frac{1}{n} D\left(P_{\tilde{M} \tilde{Y}^n} \| P_{\tilde{M}} P_{\tilde{Y}^n}\right) + \frac{2}{n} \log \Delta_n && (133) \\ &= \frac{1}{n} I(\tilde{M}; \tilde{Y}^n) + \frac{2}{n} \log \Delta_n && (134) \\ &= \frac{1}{n} \sum_{t=1}^n I(\tilde{M}; \tilde{Y}_t | \tilde{Y}^{t-1}) + \frac{2}{n} \log \Delta_n && (135) \\ &\leq I(\tilde{M}, \tilde{Y}^{T-1}, \tilde{X}^{T-1}; \tilde{Y}_T | T) + \frac{2}{n} \log \Delta_n && (136) \\ &\stackrel{(g)}{\leq} I(\tilde{V}; \tilde{Y}_T) + o(1), && (137) \end{aligned}$$

where (f) holds because $P_{\tilde{Y}^n}(y^n) \leq P_Y^{\otimes n}(y^n) \Delta_n^{-1}$ and $P_{\tilde{\mathsf{M}}}(\mathsf{m}) \leq P_{\mathsf{M}}(\mathsf{m}) \Delta_n^{-1}$, and (g) holds by the definition of the auxiliary random variable V in (127) and by the limit (125).

Bounding the Rate:

To bound the rate, notice:

$$\begin{aligned} R &\geq \frac{1}{n} H(\tilde{M}) && (138) \\ &= \frac{1}{n} I(\tilde{M}; \tilde{X}^n, \tilde{Y}^n) && (139) \\ &= \frac{1}{n} H(\tilde{X}^n, \tilde{Y}^n) - \frac{1}{n} \sum_{t=1}^n H(\tilde{X}_t, \tilde{Y}_t | \tilde{X}^{t-1}, \tilde{Y}^{t-1} \tilde{M}) && (140) \\ &\stackrel{(i)}{=} H(\tilde{X}_T, \tilde{Y}_T) + o(1) - H(\tilde{X}_T, \tilde{Y}_T | \tilde{V}) && (141) \\ &= I(\tilde{X}_T, \tilde{Y}_T; \tilde{V}) + o(1) && (142) \\ &\geq I(\tilde{X}_T; \tilde{V}) + o(1), && (143) \end{aligned}$$

where Step (i) holds by Lemma 1 and the definition of the random variable V in (127).

Approximate Markov Chain: Following the bounding steps in (58)–(62), but replacing $\tilde{U}$ by $\tilde{V}$ (see (127)), we obtain:

$$I(\tilde{V};\tilde{Y}_T|\tilde{X}_T) = o(1). \tag{144}$$

Asymptotic Analysis: Analogous to the asymptotic analysis in Subsection IV-A. ■

Remark 6: Extensions of Theorem 8 to the expected-rate constraint in (20) or to a secrecy constraint can be obtained using similar proof steps, see [3], [24]. Under an expected-rate constraint it suffices to modify the rate-bounding steps by first applying the steps used to derive (40) before continuing to bound the entropy $H(\tilde{\mathsf{M}})$ as in (139)–(143).

Change of measure proof techniques can also be use to derive tight converses for distributed testing against independence over noisy channels [25], in multi-hop and multi-decision center scenarios [1], [2], [4], [5], [26], or in zero-rate sequential testing [27].

VI. Channel Coding

This section studies communication over a discrete memoryless channel (DMC) as depicted in Figure 2. A transmitter (Tx) communicates a Message M to a receiver (Rx) over a DMC parametrized by the finite input and output alphabets $\mathcal{X}$ and $\mathcal{Y}$ and the transition law $P_{Y|X}$. Message M is uniformly distributed over the set $\mathcal{M} := \{1,\dots,2^{nR}\}$ with $R>0$ and $n>0$ denoting the rate and blocklength of communication, respectively.

The Tx produces channel inputs

$$X^n = \phi^{(n)}(M) \tag{145}$$

for some encoding function $\phi^{(n)}\colon \mathcal{M} \to \mathcal{X}^n$. For ease of notation we also write $x^n(m)$ instead of $\varphi^{(n)}(m)$.

The Rx observes the sequence of channel outputs Y^n, where Y_t is distributed according to the law $P_{Y|X}(\cdot|x)$ if $X_t = x$.

Fig. 2: Channel coding over a DMC.

Based on Y^n, the Rx produces the guess

$$\hat{M} = g^{(n)}(Y^n) \tag{146}$$

so as to minimize the error probability

$$p^{(n)}(\text{error}) := \Pr\left[\hat{M} \neq M\right]. \tag{147}$$

Definition 7: The rate $R>0$ is said ϵ-achievable over the DMC $(\mathcal{X},\mathcal{Y},P_{Y|X})$, if there exists a sequence $\{(\phi^{(n)},g^{(n)})\}_{n=1}^{\infty}$ such that

$$\overline{\lim_{n\to\infty}}\, p^{(n)}(\text{error}) \leq \epsilon. \tag{148}$$

Wolfowitz [28] established the following strong converse theorem, which implies that for all rates above capacity the probability of communication error over a discrete-memoryless channel (DMC) (see Figure 2) tends to 1 as the blocklength increases. Arimoto established the corresponding exponentially-strong converse [29] stating that the convergence takes place exponentially fast in the blocklength, was first established by Arimoto and subsequently refined by Csiszár and Körner [10] who provided lower bounds on the error exponents at rates above capacity. Since then, various alternative proofs have been presented [10], [29]–[32], as well as extensions to multi-user communication networks [33]–[35].

Theorem 9 ([28], [29]): Any rate $R>C$, where C denotes the capacity

$$C := \max_{P_X} I(X;Y), \tag{149}$$

is not ϵ-achievable for any $\epsilon \in [0,1)$.

Proof: Fix a sequence $\{(\phi^{(n)},g^{(n)})\}_{n=1}^{\infty}$ satisfying (148).

Change of Measure: Define the set

$$\mathcal{D}_n := \left\{(m,y^n)\colon y^n \in \mathcal{T}_{\mu_n}^{(n)}(P_{Y|X=x^n(m)}) \text{ and } g^{(n)}(y^n) = m\right\} \tag{150}$$

and its probability

$$\Delta_n := \Pr[(M,Y^n) \in \mathcal{D}_n]. \tag{151}$$

By now standard arguments (see (29)) and by (148):

$$\lim_{n\to\infty} \frac{1}{n}\log \Delta_n = 0. \tag{152}$$

Let further $(\tilde{M},\tilde{X}^n,\tilde{Y}^n)$ be random variables so that

$$P_{\tilde{M}\tilde{X}\tilde{Y}^n}(m,x^n,y^n) = \frac{1}{2^{nR}} \cdot \frac{P_{Y|X}^{\otimes n}(y^n|\phi^n(m))}{\Delta_n} \cdot \mathbb{1}\{(m,y^n)\in\mathcal{D}_n\}\mathbb{1}\left\{x^n = \phi^{(n)}(m)\right\} \tag{153}$$

and the time-index T as usual.

Bounding the Rate:

By definition, $P_{\tilde{M}}(m) \leq 2^{-nR}\Delta_n^{-1}$ and thus:

$$\frac{1}{n}H(\tilde{M}) \geq R + \frac{1}{n}\log\Delta_n. \tag{154}$$

We continue to notice:

$$\begin{aligned} R &\leq \frac{1}{n}H(\tilde{M}) - \frac{1}{n}\log\Delta_n && (155)\\ &\overset{(a)}{=} \frac{1}{n}I(\tilde{M};\tilde{Y}^n) - \frac{1}{n}\log\Delta_n && (156)\\ &= \frac{1}{n}H(\tilde{Y}^n) - \frac{1}{n}H(\tilde{Y}^n|\tilde{M}) - \frac{1}{n}\log\Delta_n && (157)\\ &= H(\tilde{Y}_T) + o(1) - \frac{1}{n}H(\tilde{Y}^n|\tilde{M}) - \frac{1}{n}\log\Delta_n, && (158) \end{aligned}$$

where (a) holds because by definition $\tilde{M} = g^{(n)}(\tilde{Y}^n)$.

Define P_X as any convergence point of the expected type-vector $\mathbb{E}_{\tilde{M}}[\pi_{x^n(\tilde{M})}]$ and $\{n_i\}_{i=1}^{\infty}$ the corresponding subsequence so that for any $x\in\mathcal{X}$:

$$\lim_{i\to\infty} \mathbb{E}_{\tilde{M}}[\pi_{x^{n_i}(\tilde{M})}(x)] = P_X(x), \quad x\in\mathcal{X}. \tag{159}$$

Then following lemma is proved in [1, Lemma 3]:

Lemma 3: We have

$$\lim_{i\to\infty} P_{\tilde{Y}_T}(y) = \sum_{x\in\mathcal{X}} P_X(x)P_{Y|X}(y|x) \tag{160}$$

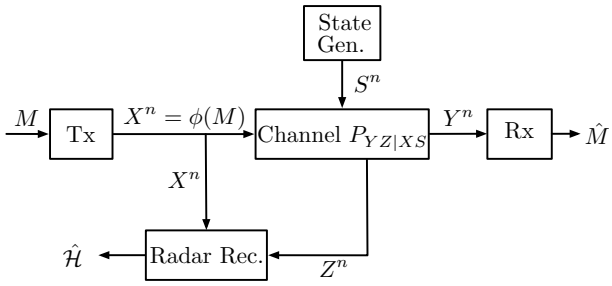

Fig. 3: Bistatic Radar ISAC Model

$$\lim_{i\to\infty} \frac{1}{n_i} H(\tilde{Y}^{n_i}|\tilde{M}) = H_{P_X P_{Y|X}}(Y|X), \tag{161}$$

where $H_{P_X P_{Y|X}}(Y|X)$ denotes the conditional entropy of Y given X when the pair $(X,Y) \sim P_X P_{Y|X}$.

Asymptotic Analysis: Restrict (158) to above chosen subsequence of blocklengths $\{n_i\}_{i=1}^{\infty}$ and plug in Equations (152) and (161). Taking $i \to \infty$, by continuity of entropy, we deduce that

$$R \le I_{P_X P_{Y|X}}(X;Y) \le C, \tag{162}$$

where the subscript indicates that mutual information is with respect to the joint pmf $P_X P_{Y|X}$. This concludes the proof of the strong converse for channel coding. ■

VII. Integrated Sensing and Communication

Consider the bistatic radar receiver model over a memoryless channel in Fig. 3, with encoding $\phi^{(n)}$ and decoding $g^{(n)}$ as described in the preceding Section VI. In this section, however the channel is governed by a state-sequence S^n whose distribution depends on a binary hypothesis:

$$\text{if } \mathcal{H}=0: \quad S^n \text{ i.i.d. } \sim P_S; \tag{163a}$$
$$\text{if } \mathcal{H}=1: \quad S^n \text{ i.i.d. } \sim Q_S, \tag{163b}$$

Based on the intpus X^n and the state sequence S^n, the channel produces the output sequence Y^n to the Rx and an additional backscattered signal Z^n to an additional *radar Rx*. At each time t the pair (Y_t, Z_t) is produced according to the memoryless and stationary channel law $P_{YZ|XS}$ based on the time-t input and state pair (X_t, S_t).

The radar Rxs attempts to guess the underlying hypothesis based on the inputs and backscattered signals:

$$\hat{\mathcal{H}} = h^{(n)}(X^n, Z^n) \in \{0,1\}. \tag{164}$$

In this context we have the following definition and result.

Definition 8: A rate-exponent pair (R,θ) is (ϵ,δ)-achievable, if there exists a sequence $\{(\phi^{(n)}, g^{(n)}, h^{(n)})\}$ so that the following three conditions hold:

$$\overline{\lim_{n\to\infty}} \Pr[\hat{M} \neq M | \mathcal{H} = \mathsf{H}, M = m] \le \epsilon, \quad \mathsf{H} \in \{0,1\}, \quad m \in \mathcal{M}, \tag{165a}$$
$$\overline{\lim_{n\to\infty}} \Pr[\hat{\mathcal{H}} = 1 | \mathcal{H} = 0] \le \delta, \tag{165b}$$
$$-\underline{\lim_{n\to\infty}} \frac{1}{n} \log \Pr[\hat{\mathcal{H}} = 0 | \mathcal{H} = 1] \ge \theta. \tag{165c}$$

For simplicity, in this section we apply the maximum decoding-error probability criterion (165a) instead of the average error probability criterion that we used in the previous section. The result and proof can be strengthened to apply also to the latter criterion as explained in [6].

Theorem 10 ([6]): For any $\epsilon, \delta \ge 0$ satisfying

$$\epsilon + \delta < 1, \tag{166}$$

a rate-exponent pair (R,θ) is not (ϵ,δ)-achievable if for all pmfs P_X

$$R < \min\{I_{P_X P_{Y|X}}(X;Y),\ I_{P_X Q_{Y|X}}(X;Y)\}, \tag{167}$$

or

$$\theta > \mathbb{E}_{P_X}\left[D(P_{Z|X} \| Q_{Z|X})\right] \tag{168}$$

where $P_{Y|X}$ and $P_{Z|X}$ denote the conditional marginals of $P_S P_{YZ|XS}$ and similarly for $Q_{Y|X}$ and $Q_{Z|X}$.

Proof: Let $\{(\phi^{(n)}, g^{(n)}, h^{(n)})\}_{n=1}^{\infty}$ satisfy (165). Let P_X denote a convergence point of the expected type-vector $\mathbb{E}_M[\boldsymbol{\pi}_{x^n(M)}]$, where we again use $x^n(m)$ to denote $\phi^{(n)}(m)$, and let $\{n_i\}_{i=1}^{\infty}$ denote the corresponding subsequence of blocklengths.

First Change of Measure: Fix $m \in \mathcal{M}$ and consider the conditions

$$g^{(n)}(y^n) = m \tag{169a}$$
$$h^{(n)}(x^n(m), z^n)) = 0 \tag{169b}$$
$$\left|\pi_{x^n(m)y^n z^n}(a,b,c) - \pi_{x^n(m)}(a) P_{YZ|X}(b,c|a)\right| \le \mu_n, \tag{169c}$$

where $P_{YZ|X}(y,z|x) \triangleq \sum_s P_S(s) P_{YZ|XS}(y,z|x,s)$. Define the set

$$\mathcal{D}_{n,m} \triangleq \{(m, y^n, z^n) \colon \quad \text{(169a)} - \text{(169c) hold}\}, \tag{170}$$

and introduce the new random variables $(\tilde{Y}^n, \tilde{Z}^n)$ of joint conditional pmf

$$P_{\tilde{Y}^n \tilde{Z}^n | \tilde{M}}(y^n, z^n m) = \frac{P_{YZ|X}^{\otimes n}(y^n, z^n | x^n(m))}{\Delta_{n,m}} \cdot \mathbb{1}\{(y^n, z^n) \in \mathcal{D}_{n,m}\} \tag{171}$$

where

$$\Delta_{n,m} \triangleq \sum_{s^n, y^n, z^n} P_{YZ|XS}^{\otimes n}(y^n, z^n | x^n(m), s^n) P_S^{\otimes n}(s^n) \cdot \mathbb{1}\{(y^n, z^n) \in \mathcal{D}_{n,m}\}. \tag{172}$$

By [10, Remark to Lemma 2.12], Condition (165), and the union bound, for each $m \in \mathcal{M}$:

$$\Delta_{n,m} \ge 1 - \frac{|\mathcal{Z}||\mathcal{Y}|}{4\mu_n^2 n} - \epsilon - \delta, \tag{173}$$

and thus by Assumption (166):

$$\frac{1}{n} \log \Delta_n = 0. \tag{174}$$

First Bound on the Rate: In the same way as (162) was proved in the previous Section VI, we can prove that

$$R \le I_{P_X P_S P_{Y|XS}}(X;Y), \tag{175}$$

Bounding the Miss-Detection Error Exponent:

Following similar steps as used to derive (132) and (133), and setting $X^n = x^n(M)$, one can show that

$$-\frac{1}{n} \log \Pr[\hat{\mathcal{H}} = 0 | \mathcal{H} = 1]$$

$$= -\frac{1}{n} \log P_{X^n} Q_{Z|X}^{\otimes n}(\hat{\mathcal{H}} = 0) \tag{176}$$

$$= \frac{1}{n} D\left(P_{X^n} P_{\tilde{Z}^n|X^n}(\hat{\mathcal{H}}) \parallel P_{X^n} Q_{Z|X}^{\otimes n}(\hat{\mathcal{H}})\right) \tag{177}$$

$$\leq \frac{1}{n} D\left(P_{X^n} P_{\tilde{Z}^n|X^n} \parallel P_{X^n} Q_{Z|X}^{\otimes n}\right) \tag{178}$$

$$= \frac{1}{n} \mathbb{E}_{P_{\tilde{X}^n}}\left[D\left(P_{\tilde{Z}^n|\tilde{X}^n} \parallel Q_{Z|X}^{\otimes n}\right)\right] \tag{179}$$

$$\leq \sum_{x,z} P_{X_T}(x) P_{\tilde{Z}_T|X_T}(z|x) \log \frac{P_{Z|X}(z|x)}{Q_{Z|X}(z|x)} - \frac{1}{n} \log \Delta_n, \tag{180}$$

for T uniform and independent. By (180) and Condition (169c) and by considering the subsequence of blocklengths $\{n_i\}$ (for which $P_{\tilde{X}_T(x)}$ converges to $P_X(x)$), we obtain the bound:

$$\overline{\lim_{i\to\infty}} -\frac{1}{n_i} \log \Pr[\hat{\mathcal{H}} = 0 | \mathcal{H} = 1] \leq \mathbb{E}_{P_X}\left[D(P_{Z|X} \| Q_{Z|X})\right]. \tag{181}$$

Second Change of Measure and Rate Bound: Repeat the first change of measure and rate bound, but where Condition (169c) is replaced by

$$\left|\pi_{x^n(m)y^n z^n}(a,b,c) - \pi_{x^n(m)}(a) Q_{YZ|X}(b,c|a)\right| \leq \mu_n, \tag{182}$$

for $Q_{YZ|X}(y,z|x) \triangleq \sum_s Q_S(s) P_{YZ|XS}(y,z|x,s)$. (Condition (169b) can either be omitted or kept as is.) This leads to the bound:

$$R \leq I_{P_X Q_S P_{Y|XS}}(X;Y). \tag{183}$$

Concluding the Proof: Combine (175), (181), and (183). ■

VIII. Conclusion and Discussion

We reviewed change of measure proof techniques for source coding, channel coding, and hypothesis testing problems. In particular we showed how this proof method can be applied to account for multiple constraints (multiple distortion constraints or mixed decoding and detection error constraints) or for expectation constraints such as expected-rate or equivocation constraints.

References

[1] M. Hamad, M. Wigger, and M. Sarkiss, "Strong converses using typical changes of measures and asymptotic Markov chains." [Online]. Available: https://arxiv.org/abs/2301.06289.pdf, 2023.

[2] M. Hamad, M. Wigger, and M. Sarkiss, "Multi-hop network with multiple decision centers under expected-rate constraints," *IEEE Transactions on Information Theory*, vol. 69, no. 7, pp. 4255–4283, 2023.

[3] S. Faour, M. Hamad, M. Sarkiss, and M. Wigger, "Testing against independence with an eavesdropper," in *2023 IEEE Information Theory Workshop (ITW)*, pp. 277–282, 2023.

[4] M. Hamad, M. Wigger, and M. Sarkiss, "Cooperative multi-sensor detection under variable-length coding," in *2020 IEEE Information Theory Workshop (ITW)*, pp. 1–5, 2021.

[5] M. Hamad, M. Sarkiss, and M. Wigger, "Benefits of rate-sharing for distributed hypothesis testing," in *2022 IEEE International Symposium on Information Theory (ISIT)*, pp. 2714–2719, 2022.

[6] M. Ahmadipour, M. Wigger, and S. Shamai, "Strong converses for memoryless bi-static ISAC," in *2023 IEEE International Symposium on Information Theory (ISIT)*, pp. 1818–1823, 2023.

[7] W. Gu and M. Effros, "A strong converse for a collection of network source coding problems," in *2009 IEEE International Symposium on Information Theory*, pp. 2316–2320, 2009.

[8] W. Gu and M. Effros, "A strong converse in source coding for super-source networks," in *2011 IEEE International Symposium on Information Theory Proceedings*, pp. 395–399, IEEE, 2011.

[9] H. Tyagi and S. Watanabe, "Strong converse using change of measure arguments," *IEEE Trans. Inf. Theory*, vol. 66, no. 2, pp. 689–703, 2019.

[10] I. Csiszár and J. Körner, *Information theory: coding theorems for discrete memoryless systems*. Cambridge University Press, 2011.

[11] D. Slepian and J. Wolf, "Noiseless coding of correlated information sources," *IEEE Transactions on Information Theory*, vol. 19, no. 4, pp. 471–480, 1973.

[12] Y. Oohama and T. S. Han, "Universal coding for the Slepian-Wolf data compression system and the strong converse theorem," *IEEE Transactions on Information Theory*, vol. 40, no. 6, pp. 1908–1919, 1994.

[13] A. I. Khinchin, "The entropy concept in probability theory," *Usp. Mat. Nay*, vol. 8, pp. 3–20, 1953.

[14] B. McMillan, "The basic theorems of information theory," *Ann. Math. Stat.*, vol. 24, pp. 196–219, 1953.

[15] I. Csiszár and G. Longo, "On error exponent for source coding and for testing simple statistical hypotheses," *Studia Sci. Math. Hungar.*, pp. 181–191, 1977.

[16] A. Wyner and J. Ziv, "The rate-distortion function for source coding with side information at the decoder," *IEEE Transactions on Information Theory*, vol. 22, no. 1, pp. 1–10, 1976.

[17] Y. Oohama, "Exponential strong converse for source coding with side information at the decoder," *Entropy*, vol. 20, no. 5, p. 352, 2018.

[18] J. Körner, "Coding of an information source having ambiguous alphabet and the entropy of graphs," in *Transactions of the 6th Prague conference on Information Theory, etc.*, (Prague, Czech Republic), pp. 411–425, 1973.

[19] A. El Gamal and Y. H. Kim, *Network Information Theory*. Cambridge University Press, 2011.

[20] O. Kosut and J. Kliewer, "Strong converses are just edge removal properties," *IEEE Transactions on Information Theory*, vol. 65, no. 6, pp. 3315–3339, 2018.

[21] L. Zhou and A. Hero, "Exponential strong converse for successive refinement with causal decoder side information," *Entropy*, vol. 21, no. 4, 2019.

[22] J. Kieffer, "Strong converses in source coding relative to a fidelity criterion," *IEEE Transactions on Information Theory*, vol. 37, no. 2, pp. 257–262, 1991.

[23] R. Ahlswede and I. Csiszár, "Hypothesis testing with communication constraints," *IEEE Trans. Inf. Theory*, vol. 32, pp. 533–542, Jul. 1986.

[24] S. Salehkalaibar and M. Wigger, "Distributed hypothesis testing with variable-length coding," *IEEE Journal on Selected Areas in Information Theory*, vol. 1, no. 3, pp. 681–694, 2020.

[25] S. Sreekumar and D. Gündüz, "Strong converse for testing against independence over a noisy channel," pp. 1283–1288, June 2020.

[26] D. Cao, L. Zhou, and V. Y. F. Tan, "Strong converse for hypothesis testing against independence over a two-hop network," *Entropy (Special Issue on Multiuser Information Theory II)*, vol. 21, Nov. 2019.

[27] S. Salehkalaibar and V. Y. Tan, "Distributed sequential hypothesis testing with zero-rate compression," in *2021 IEEE Information Theory Workshop (ITW)*, pp. 1–5, IEEE, 2021.

[28] J. Wolfowitz, "The coding of messages subject to chance errors," *Illinois Journal of Mathematics*, vol. 1, no. 4, pp. 591–606, 1957.

[29] S. Arimoto, "On the converse to the coding theorem for discrete memoryless channels (corresp.)," *IEEE Transactions on Information Theory*, vol. 19, no. 3, pp. 357–359, 1973.

[30] V. Strassen, "Asymptotische Abschätzungen in Shannons Informationstheorie," (Prague, Czech Replublic), pp. 689–723, 962.

[31] H. V. P. Y. Polyanskiy and S. Verdú, "Channel coding rate in the finite blocklength regime," *IEEE Transactions on Information Theory*, vol. 56, no. 5, pp. 2307–2359, 2010.

[32] R. Graczyk, "An elementary proof of the strong converse of the channel coding theorem." [Online]. Available: https://perso.telecom-paristech.fr/rgraczyk/documents/papers/Robert_Graczyk_An_Elementary_Proof_of_the_Strong_Converse_to_the_Channel_Coding_Theorem.pdf.

[33] G. Dueck, "The strong converse coding theorem for the multiple-access channel," *Journal of Combinatorics, Information & System Sciences*, vol. 6, no. 3, pp. 187–196, 1981.

[34] S. L. Fong and V. Y. Tan, "A proof of the strong converse theorem for gaussian multiple access channels," *IEEE Transactions on Information Theory*, vol. 62, no. 8, pp. 4376–4394, 2016.

[35] S. L. Fong and V. Y. Tan, "A proof of the strong converse theorem for Gaussian broadcast channels via the Gaussian Poincaré inequality," *IEEE Transactions on Information Theory*, vol. 63, no. 12, pp. 7737–7746, 2017.

Appendix A
Proof of Lemma 1

The second equalities in (8)–(10) follow directly from (7) and continuity of entropy. To prove (7), notice that

$$P_{\tilde{X}_T\tilde{Y}_T}(x,y) = \frac{1}{n}\sum_{t=1}^{n} P_{\tilde{X}_t\tilde{Y}_t}(x,y) \quad (184)$$
$$= \mathbb{E}\left[\frac{1}{n}\sum_{t=1}^{n}\mathbb{1}\{\tilde{X}_t = x, \tilde{Y}_t = y\}\right] \quad (185)$$
$$= \mathbb{E}[\pi_{\tilde{X}^n\tilde{Y}^n}(x,y)], \quad (186)$$

where the expectations are with respect to the tuples $\tilde{X}^n$ and $\tilde{Y}^n$. Since by the definition of the typical set,

$$|\pi_{\tilde{X}^n\tilde{Y}^n}(x,y) - P_{XY}(x,y)| \leq \mu_n, \quad (187)$$

we conclude that as $n \to \infty$ the probability $P_{\tilde{X}_T\tilde{Y}_T}(x,y)$ tends to $P_{XY}(x,y)$.

We next prove the asymptotic equality in (11). We notice

$$\frac{1}{n}D(P_{\tilde{X}^n\tilde{Y}^n}\|Q_{XY}^{\otimes n})$$
$$= \frac{1}{n}\sum_{(x^n,y^n)\in\mathcal{D}_n} P_{\tilde{X}^n\tilde{Y}^n}(x^n,y^n)\log\frac{P_{\tilde{X}^n\tilde{Y}^n}(x^n,y^n)}{Q_{XY}^{\otimes n}(x^n,y^n)} \quad (188)$$
$$\leq \frac{1}{n}\sum_{(x^n,y^n)\in\mathcal{D}_n} P_{\tilde{X}^n\tilde{Y}^n}(x^n,y^n)\log\frac{P_{XY}^{\otimes n}(x^n,y^n)}{Q_{XY}^{\otimes n}(x^n,y^n)} \quad (189)$$
$$-\frac{1}{n}\sum_{(x^n,y^n)\in\mathcal{D}_n} P_{\tilde{X}^n\tilde{Y}^n}(x^n,y^n)\log\Delta_n \quad (190)$$
$$= \frac{1}{n}\sum_{i=1}^{n}\sum_{(x,y)} P_{\tilde{X}_i\tilde{Y}_i}(x,y)\log\frac{P_{XY}(x,y)}{Q_{XY}(x,y)} - \frac{1}{n}\log\Delta_n. \quad (191)$$
$$= D(P_{XY}\|Q_{XY}) + o(1) \quad (192)$$

where the last equality holds by (7), by continuity, and by Assumption (5). This establishes (11).

To prove (8), notice:

$$\frac{1}{n}H(\tilde{X}^n,\tilde{Y}^n) + \frac{1}{n}D(P_{\tilde{X}^n\tilde{Y}^n}\|P_{XY}^{\otimes n})$$
$$= -\frac{1}{n}\sum_{(x^n,y^n)\in\mathcal{D}_n} P_{\tilde{X}^n\tilde{Y}^n}(x^n,y^n)\log P_{XY}^{\otimes n}(x^n,y^n) \quad (193)$$
$$= -\frac{1}{n}\sum_{t=1}^{n}\sum_{(x^n,y^n)\in\mathcal{D}_n} P_{\tilde{X}^n\tilde{Y}^n}(x^n,y^n)\log P_{XY}(x_t,y_t) \quad (194)$$
$$= -\frac{1}{n}\sum_{t=1}^{n}\sum_{(x_t,y_t)\in\mathcal{X}\times\mathcal{Y}} P_{\tilde{X}_t\tilde{Y}_t}(x_t,y_t)\log P_{XY}(x_t,y_t) \quad (195)$$
$$= -\sum_{(x,y)\in\mathcal{X}\times\mathcal{Y}}\left(\frac{1}{n}\sum_{t=1}^{n}P_{\tilde{X}_t\tilde{Y}_t}(x,y)\right)\log P_{XY}(x,y) \quad (196)$$
$$= -\sum_{(x,y)\in\mathcal{X}\times\mathcal{Y}} P_{\tilde{X}_T\tilde{Y}_T}(x,y)\log P_{XY}(x,y) \quad (197)$$
$$= H(\tilde{X}_T,\tilde{Y}_T) + D(P_{\tilde{X}_T\tilde{Y}_T}\|P_{XY}), \quad (198)$$

where (195) holds by the law of total probability applied to the random variables $\tilde{X}^{t-1}, \tilde{X}_{t+1}^n, \tilde{Y}^{t-1}, \tilde{Y}_{t+1}^n$. The proof of (11) is then concluded by recalling (11) and noting that Limit (7) implies that $D(P_{\tilde{X}_T\tilde{Y}_T}\|P_{XY}), \to 0$. To prove (9), notice that by the same arguments as we concluded (8), we have

$$\frac{1}{n}H(\tilde{Y}^n) + \frac{1}{n}D(P_{\tilde{Y}^n}\|P_Y^{\otimes n}) = H(\tilde{Y}_T) + D(P_{\tilde{Y}_T}\|P_Y) \quad (199)$$

and

$$\frac{1}{n}D(P_{\tilde{Y}^n}\|P_Y^{\otimes n}) \to 0 \quad (200)$$
$$D(P_{\tilde{Y}_T}\|P_Y) \to 0, \quad (201)$$

thus implying (9).

The last limit (9) follows by the chain rule and Limits (8) and (9). This concludes the proof.

Appendix B
Proof of Equation (206)

By the chain rule of entropy and because conditioning cannot increase entropy:

$$\frac{1}{n}H(\tilde{Y}^n|\tilde{X}^n,\tilde{\mathsf{M}}) = \frac{1}{n}\sum_{t=1}^{n}H(\tilde{Y}_t|\tilde{Y}_{t+1}^n,\tilde{X}^n,\tilde{\mathsf{M}}) \quad (202)$$
$$\leq \frac{1}{n}\sum_{t=1}^{n}H(\tilde{Y}_t|\tilde{Y}_{t+1}^n,\tilde{X}^{t-1},\tilde{\mathsf{M}}) \quad (203)$$
$$= H(\tilde{Y}_T|\tilde{X}_T,U). \quad (204)$$

Following similar steps:

$$\frac{1}{n}H(\tilde{X}^n|\tilde{Y}^n,\tilde{\mathsf{M}}) \leq H(\tilde{X}_T|\tilde{Y}_T,U) \quad (205)$$

To prove

$$\frac{1}{n}H(\tilde{X}^n|\tilde{Y}^n,\tilde{\mathsf{M}}) \geq H(\tilde{X}_T|\tilde{Y}_T,U) + o(1), \quad (206)$$

we recall Csiszár's sum identity [10]. So, let A^n, B^n, and S be arbitrary (tuples of) random variables and T is independent thereof and uniform over $\{1,\ldots,n\}$. Then the following equality holds:

$$H(A^n|B^nS) - H(B^n|A^nS)$$
$$= n\left(H(A_T|B_TA^{T-1}B_{T+1}^nS) - H(B_T|A_TA^{T-1}B_{T+1}^nS)\right). \quad (207)$$

With this equality, we can write:

$$\frac{1}{n}H(\tilde{X}^n|\tilde{Y}^n\tilde{\mathsf{M}})$$
$$\overset{(h)}{=} \frac{1}{n}\left[H(\tilde{X}^n|\tilde{Y}^n\tilde{\mathsf{M}}) + I(\tilde{\mathsf{M}};\tilde{Y}^n|\tilde{X}^n)\right] \quad (208)$$
$$= \frac{1}{n}\left[H(\tilde{X}^n|\tilde{Y}^n\tilde{\mathsf{M}}) - H(\tilde{Y}^n|\tilde{X}^n\tilde{\mathsf{M}})\right] + \frac{1}{n}H(\tilde{Y}^n|\tilde{X}^n) \quad (209)$$
$$\overset{(i)}{=} H(\tilde{X}_T|\tilde{Y}_T,U) - H(\tilde{Y}_T|\tilde{X}_T,U) + \frac{1}{n}H(\tilde{Y}^n|\tilde{X}^n) \quad (210)$$
$$\overset{(j)}{=} H(\tilde{X}_T|\tilde{Y}_T,U) - H(\tilde{Y}_T|\tilde{X}_T,U) + H(\tilde{Y}_T|\tilde{X}_T) + o(1) \quad (211)$$
$$\overset{(k)}{\geq} H(\tilde{X}_T|\tilde{Y}_T,U) + o(1) \quad (212)$$

where:

- (h) holds because $\tilde{\mathsf{M}}$ is a function of $\tilde{X}^n$ and therefore $I(\tilde{\mathsf{M}};\tilde{Y}^n|\tilde{X}^n) = 0$;
- (i) is obtained by applying (207) to the choice $A = \tilde{X}$, $B = \tilde{Y}$, $S = \tilde{\mathsf{M}}$ and by using the definition of U in (12);
- (j) holds by Lemma 1;
- (k) holds by the non-negativity of the mutual information term $I(\tilde{Y}_T;U|\tilde{X}_T) = H(\tilde{Y}_T|\tilde{X}_T) - H(\tilde{Y}_T|\tilde{X}_T,U)$.